Topic: Basic ML

Which of the following is a regression problem?

- ✅ **Predicting the price of a house**
  - *Why:* Regression deals with predicting continuous numerical values (like price, temperature, or distance). The other options involve predicting discrete classes, which are classification problems.

Which scaler transforms features so that each feature has zero mean and unit variance?

- ✅ **StandardScaler**
  - *Why:* `StandardScaler` standardizes a feature by subtracting the mean and dividing by the standard deviation.

Which metric is most appropriate for evaluating a regression model?

- ✅ **Mean Squared Error (MSE)**
  - *Why:* MSE measures the average squared difference between predicted and actual continuous values. F1-Score, Accuracy, and Recall are exclusively classification metrics.

Topic: Cross Validation

Which variant of cross-validation is especially recommended for imbalanced classification datasets?

- ✅ **Stratified k-fold cross-validation**
  - *Why:* Stratified k-fold ensures that the proportion of classes in each fold mirrors the proportion in the overall dataset, preventing folds from missing minority class examples.

Topic: Underfitting / Overfitting

A model achieves 99% training accuracy but only 61% validation accuracy. This is most likely an example of:

- ✅ **Overfitting (high variance)**
  - *Why:* A massive gap between training performance (very high) and validation performance (poor) means the model has memorized the training data's noise rather than learning generalizable patterns.

Which pattern is a symptom of underfitting?

- ✅ **Low training accuracy and low test accuracy**
  - *Why:* If the model cannot even perform well on the data it is actively learning from, it lacks the capacity (or hasn't been trained enough) to capture the underlying relationships.

Topic: Linear & Logistic Regression

What is the output of the sigmoid function σ(z)=1+e−z1 when z = 0?

- ✅ **0.5**
  - *Why:* When z=0, e−0=1. The equation becomes 1+11=21=0.5.

In linear regression what cost function does gradient descent minimise?

- ✅ **Mean squared error (MSE)**
  - *Why:* Ordinary Least Squares (OLS) linear regression relies on minimizing the sum of the squared residuals, which equates to minimizing the MSE.

Topic: SVMs

What are support vectors in a Support Vector Machine?

- ✅ **The training examples closest to the decision boundary that define the margin**
  - *Why:* These are the critical data points that "support" the margin. If you were to remove all other data points, the position of the separating hyperplane would remain exactly the same.

Topic: PyTorch Tensors

What is the shape of the result of multiplying a tensor of shape (3, 4) with a tensor of shape (4, 5) using the @ operator?

- ✅ **(3, 5)**
  - *Why:* The `@` operator performs matrix multiplication. The inner dimensions (`4`) must match and collapse, leaving the outer dimensions `(3, 5)`.

Which PyTorch expression computes the element-wise (Hadamard) product of two 2D tensors A and B of the same shape?

- ✅ **A * B**
  - *Why:* The standard asterisk `*` operator in PyTorch performs element-wise multiplication.

Topic: Neural Networks

What is the main purpose of an activation function in a neural network?

- ✅ **To introduce non-linearity so the network can learn complex patterns**
  - *Why:* Without non-linear activation functions (like ReLU, Sigmoid, or Tanh), a neural network with dozens of layers would mathematically collapse down to a single linear transformation, losing its ability to solve complex problems.

Topic: CNN

What does max pooling do in a CNN?

- ✅ **Takes the maximum value in each pooling window to reduce spatial dimensions**
  - *Why:* Max pooling downsamples the feature maps, reducing the computational load and extracting the most prominent features (like edges) while discarding less important spatial information.

Topic: Decision Trees

A decision tree is grown to maximum depth with no stopping criterion. On training data it achieves 100% accuracy. On test data accuracy is poor. What is the most effective fix?

- ✅ **Limit the maximum depth**
  - *Why:* The tree has overfit by creating a leaf node for almost every single training instance. Limiting depth (pruning) forces the tree to generalize.

In a classification tree, how is the class prediction at a leaf node determined?

- ✅ **It is the majority class among the training examples at that leaf**
  - *Why:* Once a new data point traverses the tree and lands in a leaf node, the model looks at the training labels that ended up in that same node and simply takes a majority vote.

Topic: PyTorch Tensors (Continued)

What is the result of `torch.tensor([1.0, 2.0, 3.0]).unsqueeze(0).shape`?

- ✅ **(1, 3)**

    - *Why:* The original tensor has a shape of `(3,)`. The `.unsqueeze(0)` method inserts a new dimension of size `1` at index `0`, resulting in a shape of `(1, 3)`.

What does the `.item()` method do on a PyTorch tensor such as `loss.item()`?

- ✅ **Extracts the scalar value as a plain Python number**
    - *Why:* This is commonly used during training loops to pull the loss value out of the computation graph so it can be safely printed or logged without causing memory leaks.

Topic: Neural Networks (Continued)

What does the Softmax function ensure about its output vector?

- ✅ **Each output is between 0 and 1 AND the outputs sum to 1**
    - *Why:* Softmax converts a vector of raw scores (logits) into a valid probability distribution.

Topic: PyTorch Training

What is the main benefit of wrapping inference code in a `torch.no_grad()` context manager?

- ✅ **It disables gradient tracking, reducing memory usage and speeding up inference by not building the computation graph**
    - *Why:* During inference (or validation), you don't need to backpropagate, so saving the intermediate activations required for gradients is an unnecessary waste of memory and time.

What does setting `shuffle=True` on a PyTorch DataLoader do?

- ✅ **It randomly re-orders the training examples at the start of each epoch to reduce sensitivity to sample ordering**
    - *Why:* Shuffling breaks up sequence biases in the dataset, ensuring that each mini-batch is a representative sample and preventing the model from learning the specific order in which the data was saved.